Introduction to Lava Flows using VICTOR

Running MOLASSES on VICTOR with insights from the 2022 Mauna Loa Eruption



source: USGS

Overview of the Lab

Part 1: Familiarizing yourself with the eruption area

Part 2: Getting Started on VICTOR

Part 3: Running the Model: MOLASSES

Part 4: Recording MOLASSES Outputs

Part 5: Analyze your MOLASSES Outputs

Part 6: Share and Discuss

Lava Flows

MOLASSES on VICTOR

Objectives:

1. **Identify** and **navigate** key aspects of the VICTOR interface.
2. **Analyze** real-world data from the 2022 eruptions of Mauna Loa and **conduct** a parameter sweep using VICTOR.
3. **Compare** and **contrast** outputs from two VICTOR models.

Introduction and Background:

In this assignment, you will use the numerical flow models MOLASSES to simulate the 2022 eruption of Mauna Loa volcano in Hawai'i. You will examine the effect of input parameters on model predictions and use the models to forecast the impact of the flow on infrastructure.



source: USGS

Preparation and Materails:

you will need...

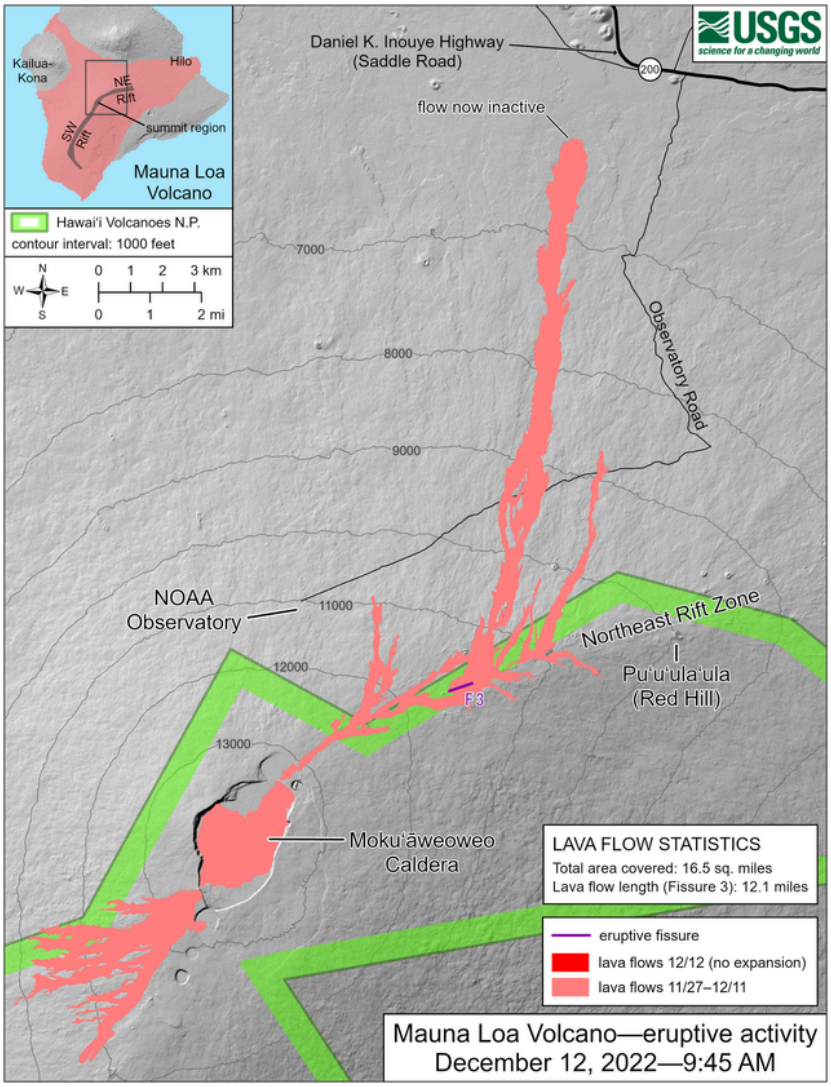
- Access to the Internet.
- To be registered on VICTOR.
  - If you have not completed this step yet, navigate to victorproject.org and fill out the form under 'sign up' in the menu.
  - ✱ The VICTOR team needs to manually confirm all registrations, so this may take up to 24 hours.

- To sign up for the VICTOR forum: forum.victorproject.org/signup
  - https://forum.victorproject.org/signup
  - Use this as a resource to ask for help or search for previously asked questions from other users.
- Optional tasks:
  - Background reading or review:
    - Molasses background: tinyurl.com/nzcjkh25
  - Watch the Introduction to VICTOR Youtube tutorial www.youtube.com/@volcanocyber

Tasks and Procedures

Part One

Familiarizing yourself with the eruption area:



Use the map to complete this section. The vent is marked in purple and labeled 'F3.'

Use the map scale, and measure and record the distance to the highway (labeled saddle road) from the vent.

Tasks and Procedures

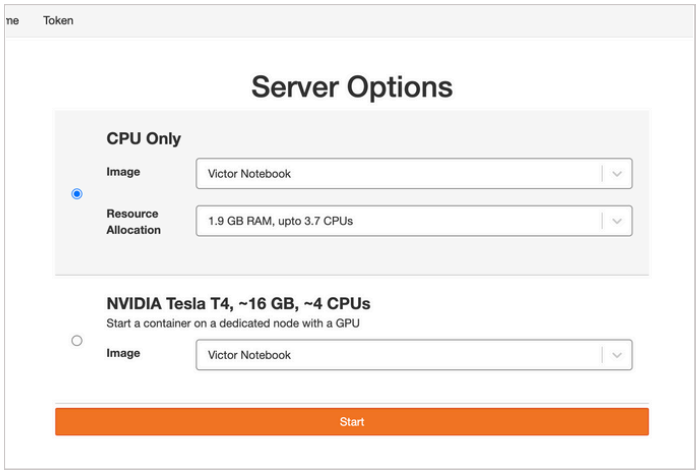
Part Two:

Getting Started on VICTOR:

1. Log in to VICTOR at hub.victorproject.org
2. **"Server Options"**
   - Ensure Victor Notebook is selected
   - Under resource allocation, select



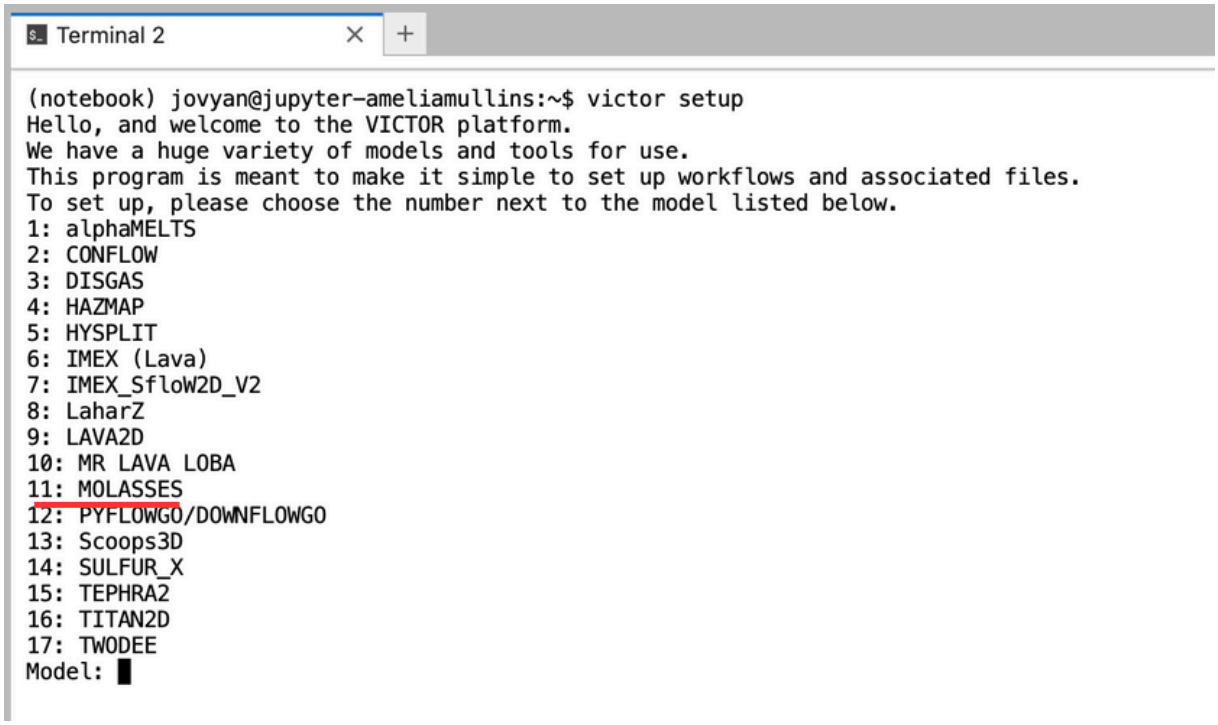
   - Clicking Start will bring you to The Launcher.



3. **The Launcher** and **VICTOR Setup command**



- From the Launcher, open a new terminal by clicking the **terminal icon** at the bottom left of your launcher.
- In the new terminal, type `victor setup` and press enter.
- Identify `MOLASSES` in the list that appears
- Type its corresponding number and press enter.



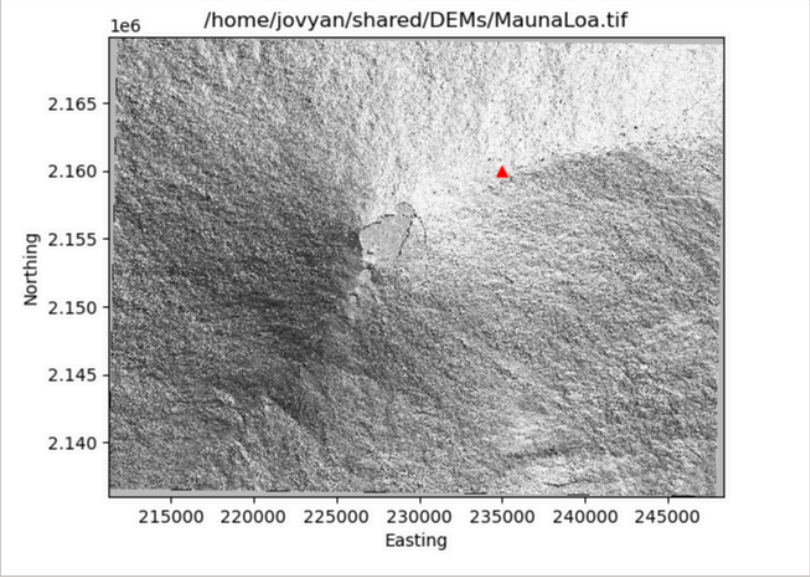
- On the left of your screen, a folder called Molasses will appear.
- Open the folder by double clicking it and find the Jupyter Notebook for molasses.ipynb

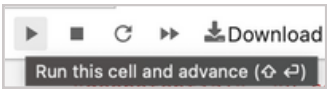
Tasks and Procedures

Part Three:

Running the Model: Molasses

- From the folder tree on the left of your screen, open the Molasses folder, and double click the molasses.ipynb notebook to open it.
- Run the notebook without changes. Press the **play button** at the top of your notebook to run each cell.
- Using the table on page 5, record the volume for this run.
- Once the cell called `Let's confirm your DEM works!` is finished, you will see a visual output similar the this:



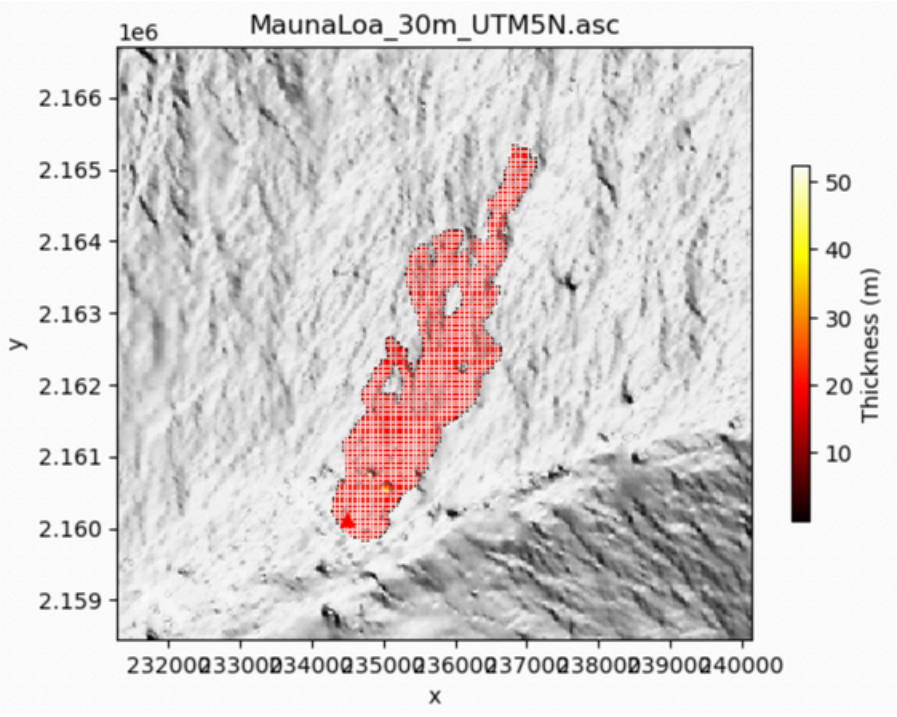


Run each cell by clicking the play button at the top of the notebook or press Shift + Enter



If the cell number is replaced with an asterisk, the cell is still running!

- Finish running the model without changes. Below is what your visual output will look like. Save the visual output as a .png for model comparison later.



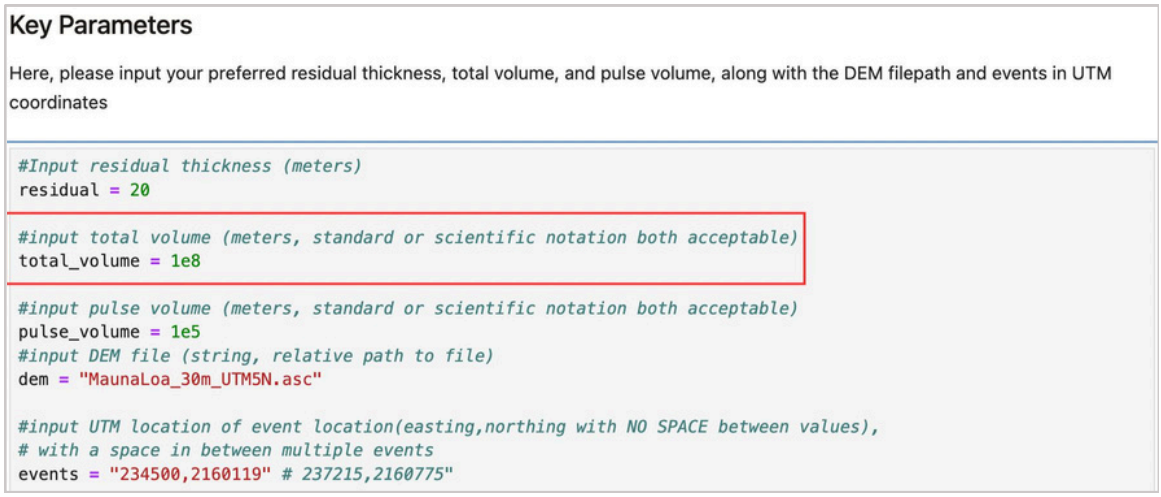
To save your visual output: Press and hold **shift**, then **right-click on the image**, then select 'save image as' to save the visual output as a .png to your device

Tasks and Procedures

Part Three:

Running the Model: Molasses (continued)

- **Next,** you will run the model again while **changing the volume to 2x the original**.
- Run the rest of the model as is, without changes.
- Using your visual output from this second run, record the input volume and flow length associated with 2x the volume in the table below.



[Changing volume parameter]

- Run the model a third time, **using 5x the original volume.** Repeat the above steps, and complete the table below with the input volume and flow length from your third output.

Part Four:

Recording Molasses Outputs

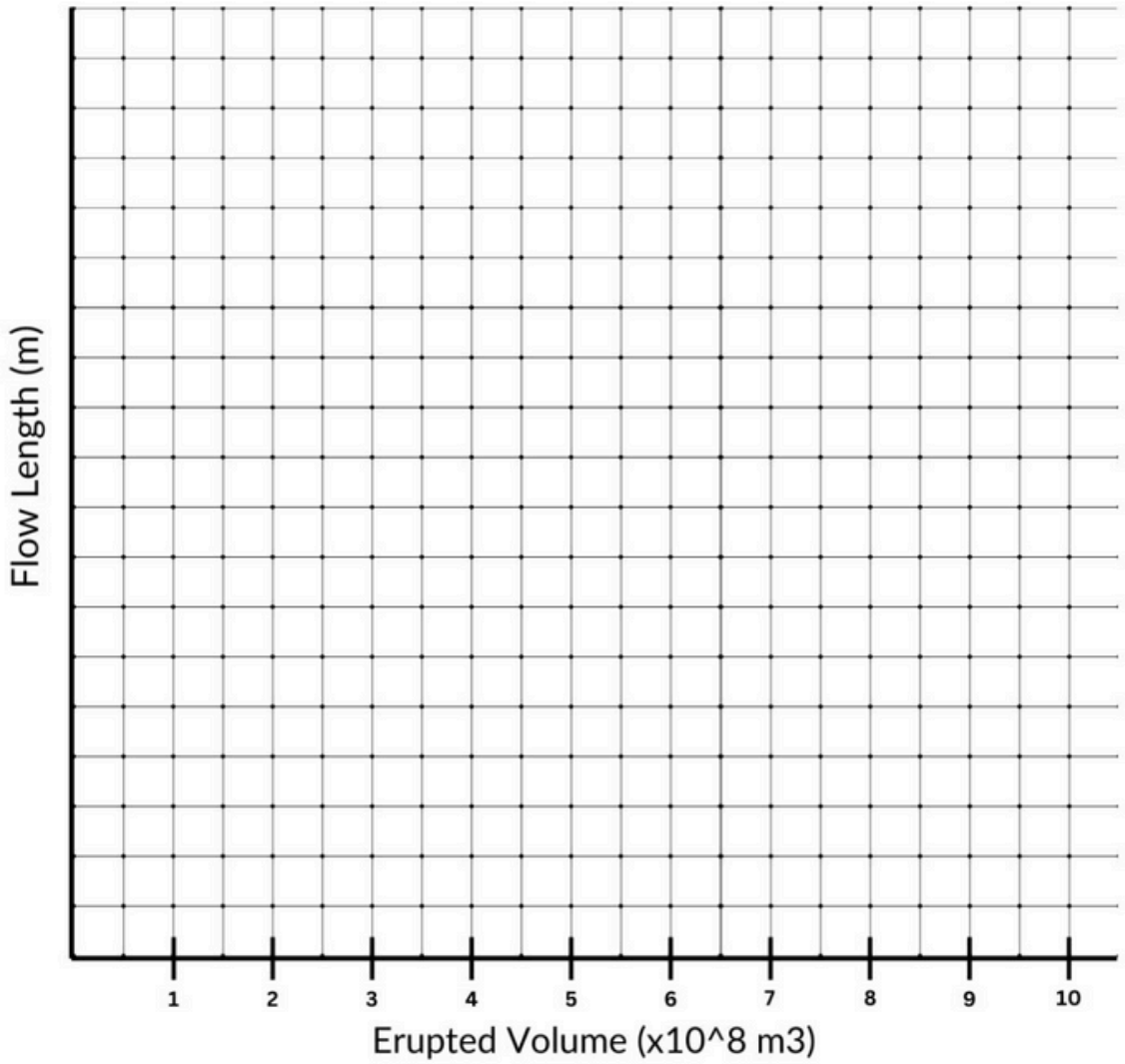
Volume	Flow length (m)	Remaining distance to highway (m)
1e8		

RECORD AND ANALYZE

Part Four:

Recording Molasses Outputs: (continued)

- Use the graph below to record and visualize your outputs:



Part Five:

Analyze Your Outputs (Molasses):

- Based on these three measurements, make a guess: what volume will be required for the flow to reach the highway according to MOLASSES?

- Run MOLASSES with this volume and test your hypothesis. Were you correct?

  yes / no

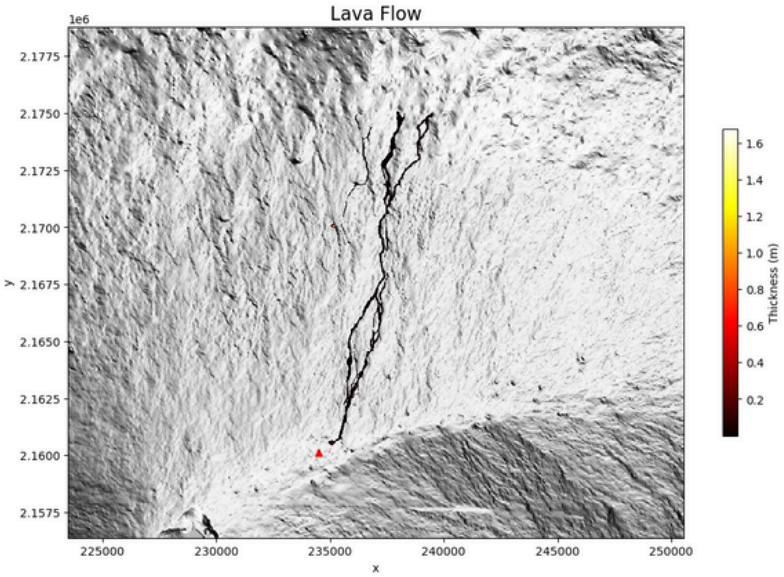
- At a constant eruption rate of 20 m3/second, how many days will it take to reach that volume?

DISCUSSION

Part Nine:

Share and Discuss

- Follow this link to find VICTOR's Lava Flow Padlet: tinyurl.com/phvyrf2d
- You will see visual outputs share from previous classes that completed this lab.
- Share you saved visual output from MOLASSES on a new note.
- In your post, **briefly assess discuss the differences in the models and compare and contrast your experience using both models. Use the visual output example from the model Mr Lava Loba, below, in your answer**



Mr Lava Loba flow output use the same Mauna Loa DEM, with a volume of 36000m3

- Compare the USGS map provided on page 2 to your models' outputs. **Reflect on and discuss how accurate the models' outputs were compared to what really happened.** Include your thoughts in your post on Padlet.

Check your work!
Did you complete all 6 parts?

- ☐ Part 1: Familiarizing yourself with the eruption area
- ☐ Part 2: Getting Started on VICTOR
- ☐ Part 3: Running the Model: MOLASSES
- ☐ Part 4: Recording MOLASSES Outputs
- ☐ Part 5: Analyze your MOLASSES Outputs
- ☐ Part 6: Share and Discuss